

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: 5/8/2018 2:00:05 PM
To: "Sneller, Cindy (DEQ)" <SNELLERC@michigan.gov>
Subject: Hauling
Attachments: IMG951507.jpg, IMG951506.jpg, Pace Biosolids 3-28-2018.pdf

Cindy,

Attached are the photos from the application site and the sludge analysis.

The field they will be applying are as follows:

1. 20N7W7-RJ01
2. 21N9W19TC01
3. 21N9W20TC01
4. 21N9W30GR01

If you need anything else feel free to contact me.

Thanks Doug

Douglas Langworthy

Assistant Water Resources Supervisor

City of Cadillac

231-775-2841

ATTACHMENT NAME:

IMG951507.jpg

ATTACHMENT TYPE:

JPEG raster lossy image



MI
MIDWEST INJECTION INC.

MUNICIPAL AGRICULTURAL
855-644-2013
www.midwestinjection.com

ATTACHMENT NAME:

IMG951506.jpg

ATTACHMENT TYPE:

JPEG raster lossy image



MII

MIDWEST INJECTION INC.

855-644-2013

www.midwestinjection.com

TRUCK COUNTY

GEA Farm Technologies
Houle

GEA

ATTACHMENT NAME:

Pace Biosolids 3-28-2018.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

April 23, 2018

Cindy Tomaszewski
City of Cadillac
200 North Lake St.
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 4610111

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 29, 2018. The results relate only to the samples included in this report. Results reported herein conform to the most current, applicable TNI/NELAC standards and the laboratory's Quality Assurance Manual, where applicable, unless otherwise noted in the body of the report.

Unreported: Removed TICs

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Andrea Grube
andrea.grube@pacelabs.com
269-381-9666
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids

Pace Project No.: 4610111

Grand Rapids Certification ID's

5560 Corporate Exchange Ct SE, Grand Rapids, MI 49512

Minnesota Department of Health, Certificate #1385941

Arkansas Department of Environmental Quality, Certificate
#17-046-0

Georgia Environmental Protection Division, Stipulation

Illinois Environmental Protection Agency, Certificate

#004325

Michigan Department of Environmental Quality, Laboratory

#0034

New York State Department of Health, Serial #56192 and
56193

North Carolina Division of Water Resources, Certificate
#659

Virginia Department of General Services, Certificate #9028

Wisconsin Department of Natural Resources, Laboratory

#999472650

U.S. Department of Agriculture Permit to Receive Soil,

Permit #P330-17-00278

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 4610111

Lab ID	Sample ID	Matrix	Date Collected	Date Received
4610111001	Holding Tank 4	Solid	03/28/18 13:40	03/29/18 12:40
4610111002	Holding Tank 4	Water	03/28/18 13:40	03/29/18 12:40

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SAMPLE ANALYTE COUNT

Project: Biosolids

Pace Project No.: 4610111

Lab ID	Sample ID	Method	Analysts	Analytes Reported
4610111001	Holding Tank 4	EPA 8081B	CAC	22
		EPA 8081B	CAC	9
		EPA 8082A	MSZ	9
		EPA 8151A	JLB1	3
		EPA 9056A	JLB1	3
		EPA 6010C	KLV	5
		EPA 6010C	KLV	8
		EPA 6020A	DSC	10
		EPA 7470A	DWJ	1
		EPA 7471B	DWJ	1
		EPA 8270C	JLB	66
		EPA 8270C	JLB	17
		EPA 8260B	DLV	15
		SM 2540 G-11/3550	NS1	1
		SM 2540G	NS1	1
		SM 2540G	NS1	1
		EPA 9045D	AMM	1
		EPA 351.2	EB1	1
		SM 4500-NH3 G-11	EB1	1
		SM 4500-P E-11	NRC	1
4610111002	Holding Tank 4	EPA 8260B	BAG	37

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4 **Lab ID: 4610111001** Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081B GCS Pesticide Solids Analytical Method: EPA 8081B Preparation Method: EPA 3545A									
Aldrin	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	309-00-2	
alpha-BHC	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	319-84-6	
beta-BHC	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	319-85-7	
delta-BHC	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	319-86-8	
gamma-BHC (Lindane)	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	58-89-9	
Chlordane (Technical)	<4150	ug/kg	4150		1	04/09/18 10:29	04/18/18 04:38	57-74-9	
alpha-Chlordane	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	5103-71-9	
gamma-Chlordane	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	5103-74-2	
4,4'-DDD	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	72-54-8	
4,4'-DDE	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	72-55-9	
4,4'-DDT	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	50-29-3	
Dieldrin	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	60-57-1	
Endosulfan I	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	959-98-8	
Endosulfan II	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	33213-65-9	
Endosulfan sulfate	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	1031-07-8	
Endrin	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	72-20-8	
Endrin aldehyde	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	7421-93-4	
Heptachlor	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	76-44-8	
Heptachlor epoxide	<1410	ug/kg	1410		1	04/09/18 10:29	04/18/18 04:38	1024-57-3	
Toxaphene	<14100	ug/kg	14100		1	04/09/18 10:29	04/18/18 04:38	8001-35-2	
Surrogates									
Decachlorobiphenyl (S)	40	%.	58-128		1	04/09/18 10:29	04/18/18 04:38	2051-24-3	S0
Tetrachloro-m-xylene (S)	44	%.	63-125		1	04/09/18 10:29	04/18/18 04:38	877-09-8	S0
8081B GCS Pesticides, TCLP Analytical Method: EPA 8081B Preparation Method: EPA 3510C									
Leachate Method/Date: EPA 1311; 04/03/18 10:10									
gamma-BHC (Lindane)	ND	mg/L	0.0050	.4	1	04/09/18 13:38	04/18/18 00:51	58-89-9	
Chlordane (Technical)	ND	mg/L	0.025	.03	1	04/09/18 13:38	04/18/18 00:51	57-74-9	
Endrin	ND	mg/L	0.0050	.02	1	04/09/18 13:38	04/18/18 00:51	72-20-8	
Heptachlor	ND	mg/L	0.0050	.008	1	04/09/18 13:38	04/18/18 00:51	76-44-8	
Heptachlor epoxide	ND	mg/L	0.0050		1	04/09/18 13:38	04/18/18 00:51	1024-57-3	
Methoxychlor	ND	mg/L	0.0050	10	1	04/09/18 13:38	04/18/18 00:51	72-43-5	CU
Toxaphene	ND	mg/L	0.10	.5	1	04/09/18 13:38	04/18/18 00:51	8001-35-2	
Surrogates									
Decachlorobiphenyl (S)	78	%.	61-115		1	04/09/18 13:38	04/18/18 00:51	2051-24-3	
Tetrachloro-m-xylene (S)	71	%.	68-111		1	04/09/18 13:38	04/18/18 00:51	877-09-8	
8082 GCS Solids ASE Analytical Method: EPA 8082A Preparation Method: EPA 3545A									
PCB-1016 (Aroclor 1016)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	12674-11-2	
PCB-1221 (Aroclor 1221)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	11104-28-2	
PCB-1232 (Aroclor 1232)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	11141-16-5	
PCB-1242 (Aroclor 1242)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	53469-21-9	
PCB-1248 (Aroclor 1248)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	12672-29-6	
PCB-1254 (Aroclor 1254)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	11097-69-1	
PCB-1260 (Aroclor 1260)	<2740	ug/kg	2740		1	04/05/18 08:26	04/07/18 06:43	11096-82-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4 **Lab ID: 4610111001** Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 GCS Solids ASE Analytical Method: EPA 8082A Preparation Method: EPA 3545A									
Surrogates									
Decachlorobiphenyl (S)	51	%.	45-135		1	04/05/18 08:26	04/07/18 06:43	2051-24-3	
Tetrachloro-m-xylene (S)	80	%.	56-123		1	04/05/18 08:26	04/07/18 06:43	877-09-8	
8151A CI Acid Herbicides TCLP Analytical Method: EPA 8151A Preparation Method: EPA 8151A Leachate Method/Date: EPA 1311; 04/03/18 10:10									
2,4-D	ND	mg/L	0.62	10	1	04/09/18 12:34	04/11/18 09:19	94-75-7	
2,4,5-TP (Silvex)	ND	mg/L	0.062	1	1	04/09/18 12:34	04/11/18 09:19	93-72-1	
Surrogates									
2,4-DCAA (S)	92	%.	33-153		1	04/09/18 12:34	04/11/18 09:19	19719-28-9	
9056 IC Anions Analytical Method: EPA 9056A Preparation Method: EPA 9056A									
Chloride	13700	mg/kg	1670		1	04/09/18 19:04	04/10/18 08:14	16887-00-6	
Nitrate as N	<83.3	mg/kg	83.3		1	04/09/18 19:04	04/10/18 08:14	14797-55-8	
Sulfate	897	mg/kg	833		1	04/09/18 19:04	04/10/18 08:14	14808-79-8	R1
6010C MET ICP Analytical Method: EPA 6010C Preparation Method: EPA 3050B									
Aluminum	3630	mg/kg	811		1	04/02/18 07:55	04/03/18 10:10	7429-90-5	
Calcium	31900	mg/kg	4060		1	04/02/18 07:55	04/03/18 10:10	7440-70-2	
Magnesium	5940	mg/kg	4060		1	04/02/18 07:55	04/03/18 10:10	7439-95-4	
Potassium	7230	mg/kg	4060		1	04/02/18 07:55	04/03/18 10:10	7440-09-7	
Sodium	11100	mg/kg	4060		1	04/02/18 07:55	04/03/18 10:10	7440-23-5	
6010C MET ICP, TCLP Analytical Method: EPA 6010C Preparation Method: EPA 3010A Leachate Method/Date: EPA 1311; 04/03/18 10:10									
Arsenic	<0.10	mg/L	0.10	5	1	04/05/18 07:59	04/09/18 12:14	7440-38-2	
Barium	0.041	mg/L	0.010	100	1	04/05/18 07:59	04/09/18 12:14	7440-39-3	
Cadmium	<0.010	mg/L	0.010	1	1	04/05/18 07:59	04/09/18 12:14	7440-43-9	
Chromium	<0.050	mg/L	0.050	5	1	04/05/18 07:59	04/09/18 12:14	7440-47-3	
Lead	<0.050	mg/L	0.050	5	1	04/05/18 07:59	04/09/18 12:14	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	1	04/05/18 07:59	04/09/18 12:14	7782-49-2	
Silver	<0.010	mg/L	0.010	5	1	04/05/18 07:59	04/09/18 12:14	7440-22-4	
Zinc	0.47	mg/L	0.020		1	04/05/18 07:59	04/09/18 12:14	7440-66-6	
6020A MET ICPMS Analytical Method: EPA 6020A Preparation Method: EPA 3050B									
Arsenic	22.8	mg/kg	8.0		1	04/02/18 07:55	04/03/18 10:16	7440-38-2	
Cadmium	<4.0	mg/kg	4.0		1	04/02/18 07:55	04/03/18 10:16	7440-43-9	
Chromium	105	mg/kg	8.0		1	04/02/18 07:55	04/03/18 10:16	7440-47-3	
Copper	846	mg/kg	40.2		5	04/02/18 07:55	04/03/18 10:22	7440-50-8	
Lead	19.3	mg/kg	8.0		1	04/02/18 07:55	04/03/18 10:16	7439-92-1	
Molybdenum	22.8	mg/kg	8.0		1	04/02/18 07:55	04/03/18 10:16	7439-98-7	
Nickel	31.1	mg/kg	8.0		1	04/02/18 07:55	04/03/18 10:16	7440-02-0	
Selenium	<8.0	mg/kg	8.0		1	04/02/18 07:55	04/03/18 11:05	7782-49-2	
Silver	<4.0	mg/kg	4.0		1	04/02/18 07:55	04/03/18 10:16	7440-22-4	
Zinc	1100	mg/kg	402		5	04/02/18 07:55	04/03/18 10:22	7440-66-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4 **Lab ID: 4610111001** Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP									
Analytical Method: EPA 7470A Preparation Method: EPA 7470A									
Leachate Method/Date: EPA 1311; 04/03/18 10:10									
Mercury	<0.00020	mg/L	0.00020		1	04/09/18 12:15	04/10/18 16:10	7439-97-6	M1
7471 Mercury									
Analytical Method: EPA 7471B Preparation Method: EPA 7471B									
Mercury	<1.2	mg/kg	1.2		1	04/10/18 13:05	04/11/18 11:44	7439-97-6	M1
8270C MSSV Solid									
Analytical Method: EPA 8270C Preparation Method: EPA 3550C									
Acenaphthene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	83-32-9	
Acenaphthylene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	208-96-8	
Anthracene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	120-12-7	
Benidine	<333000	ug/kg	333000		1	04/04/18 10:24	04/14/18 15:30	92-87-5	L2
Benzo(a)anthracene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	56-55-3	
Benzo(a)pyrene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	50-32-8	
Benzo(b)fluoranthene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	205-99-2	
Benzo(g,h,i)perylene	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	191-24-2	
Benzo(k)fluoranthene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	207-08-9	
4-Bromophenylphenyl ether	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	101-55-3	
Butylbenzylphthalate	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	85-68-7	
4-Chloro-3-methylphenol	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	59-50-7	
2-Chloroaniline	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	95-51-2	L2
4-Chloroaniline	<33300	ug/kg	33300		1	04/04/18 10:24	04/14/18 15:30	106-47-8	
bis(2-Chloroethoxy)methane	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	111-91-1	
bis(2-Chloroethyl) ether	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	111-44-4	
bis(2-Chloroisopropyl) ether	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	108-60-1	
2-Chloronaphthalene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	91-58-7	
2-Chlorophenol	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	95-57-8	
4-Chlorophenylphenyl ether	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	7005-72-3	
Chrysene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	218-01-9	
Dibenz(a,h)anthracene	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	53-70-3	
Dibenzofuran	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	132-64-9	
1,2-Dichlorobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	95-50-1	
1,3-Dichlorobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	541-73-1	
1,4-Dichlorobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	106-46-7	
3,3'-Dichlorobenzidine	<333000	ug/kg	333000		1	04/04/18 10:24	04/14/18 15:30	91-94-1	
2,4-Dichlorophenol	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	120-83-2	
Diethylphthalate	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	84-66-2	
2,4-Dimethylphenol	<84500	ug/kg	84500		1	04/04/18 10:24	04/14/18 15:30	105-67-9	
Dimethylphthalate	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	131-11-3	
Di-n-butylphthalate	<33300	ug/kg	33300		1	04/04/18 10:24	04/14/18 15:30	84-74-2	
4,6-Dinitro-2-methylphenol	<84500	ug/kg	84500		1	04/04/18 10:24	04/14/18 15:30	534-52-1	
2,4-Dinitrophenol	<84500	ug/kg	84500		1	04/04/18 10:24	04/14/18 15:30	51-28-5	
2,4-Dinitrotoluene	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	121-14-2	
2,6-Dinitrotoluene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	606-20-2	
Di-n-octylphthalate	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	117-84-0	
1,2-Diphenylhydrazine	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	122-66-7	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4 Lab ID: 4610111001 Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270C MSSV Solid Analytical Method: EPA 8270C Preparation Method: EPA 3550C									
bis(2-Ethylhexyl)phthalate	20000	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	117-81-7	
Fluoranthene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	206-44-0	
Fluorene	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	86-73-7	
Hexachloro-1,3-butadiene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	87-68-3	
Hexachlorobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	118-74-1	
Hexachlorocyclopentadiene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	77-47-4	
Hexachloroethane	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	67-72-1	
Indeno(1,2,3-cd)pyrene	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	193-39-5	
Isophorone	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	78-59-1	
Naphthalene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	91-20-3	
Nitrobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	98-95-3	
2-Nitrophenol	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	88-75-5	
4-Nitrophenol	<333000	ug/kg	333000		1	04/04/18 10:24	04/14/18 15:30	100-02-7	
N-Nitrosodimethylamine	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	62-75-9	
N-Nitroso-di-n-propylamine	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	621-64-7	
N-Nitrosodiphenylamine	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	86-30-6	
Pentachlorophenol	<16400	ug/kg	16400		1	04/04/18 10:24	04/14/18 15:30	87-86-5	
Phenanthrene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	85-01-8	
Phenol	<84500	ug/kg	84500		1	04/04/18 10:24	04/14/18 15:30	108-95-2	L2,P1
Pyrene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	129-00-0	
1,2,4-Trichlorobenzene	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	120-82-1	
2,4,6-Trichlorophenol	<8450	ug/kg	8450		1	04/04/18 10:24	04/14/18 15:30	88-06-2	

Surrogates

Nitrobenzene-d5 (S)	60	%.	33-131		1	04/04/18 10:24	04/14/18 15:30	4165-60-0	
2-Fluorobiphenyl (S)	75	%.	46-122		1	04/04/18 10:24	04/14/18 15:30	321-60-8	
o-Terphenyl (S)	67	%.	20-155		1	04/04/18 10:24	04/14/18 15:30	84-15-1	
Phenol-d6 (S)	15	%.	30-115		1	04/04/18 10:24	04/14/18 15:30	13127-88-3	S0
2-Fluorophenol (S)	40	%.	33-113		1	04/04/18 10:24	04/14/18 15:30	367-12-4	
2,4,6-Tribromophenol (S)	87	%.	12-124		1	04/04/18 10:24	04/14/18 15:30	118-79-6	

8270C MSSV TCLP Sep Funnel

Analytical Method: EPA 8270C Preparation Method: EPA 3510C

Leachate Method/Date: EPA 1311; 04/03/18 10:10

2,4-Dinitrotoluene	ND	mg/L	0.10	.13	1	04/09/18 13:45	04/14/18 02:35	121-14-2	
Hexachloro-1,3-butadiene	ND	mg/L	0.10	.5	1	04/09/18 13:45	04/14/18 02:35	87-68-3	
Hexachlorobenzene	ND	mg/L	0.10	.13	1	04/09/18 13:45	04/14/18 02:35	118-74-1	
Hexachloroethane	ND	mg/L	0.20	3	1	04/09/18 13:45	04/14/18 02:35	67-72-1	
2-Methylphenol(o-Cresol)	ND	mg/L	0.20	200	1	04/09/18 13:45	04/14/18 02:35	95-48-7	
3&4-Methylphenol(m&p Cresol)	ND	mg/L	0.40	200	1	04/09/18 13:45	04/14/18 02:35		M1
Nitrobenzene	ND	mg/L	0.20	2	1	04/09/18 13:45	04/14/18 02:35	98-95-3	
Pentachlorophenol	ND	mg/L	1.5	100	1	04/09/18 13:45	04/14/18 02:35	87-86-5	
Pyridine	ND	mg/L	1.0	5	1	04/09/18 13:45	04/14/18 02:35	110-86-1	
2,4,5-Trichlorophenol	ND	mg/L	0.60	400	1	04/09/18 13:45	04/14/18 02:35	95-95-4	
2,4,6-Trichlorophenol	ND	mg/L	0.20	2	1	04/09/18 13:45	04/14/18 02:35	88-06-2	
Surrogates									
Nitrobenzene-d5 (S)	101	%.	38-141		1	04/09/18 13:45	04/14/18 02:35	4165-60-0	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4 Lab ID: 4610111001 Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270C MSSV TCLP Sep Funnel									
Analytical Method: EPA 8270C Preparation Method: EPA 3510C									
Leachate Method/Date: EPA 1311; 04/03/18 10:10									
Surrogates									
2-Fluorobiphenyl (S)	97	%	41-132		1	04/09/18 13:45	04/14/18 02:35	321-60-8	
o-Terphenyl (S)	106	%	39-148		1	04/09/18 13:45	04/14/18 02:35	84-15-1	
Phenol-d6 (S)	80	%	10-112		1	04/09/18 13:45	04/14/18 02:35	13127-88-3	
2-Fluorophenol (S)	86	%	10-123		1	04/09/18 13:45	04/14/18 02:35	367-12-4	
2,4,6-Tribromophenol (S)	83	%	20-142		1	04/09/18 13:45	04/14/18 02:35	118-79-6	
8260B MSV TCLP									
Analytical Method: EPA 8260B Leachate Method/Date: EPA 1311; 04/03/18 12:50									
Benzene	ND	mg/L	0.10	.5	100		04/05/18 00:12	71-43-2	
2-Butanone (MEK)	ND	mg/L	2.0	200	100		04/05/18 00:12	78-93-3	
Carbon tetrachloride	ND	mg/L	0.10	.5	100		04/05/18 00:12	56-23-5	
Chlorobenzene	ND	mg/L	0.10	100	100		04/05/18 00:12	108-90-7	
Chloroform	ND	mg/L	0.10	6	100		04/05/18 00:12	67-66-3	
1,4-Dichlorobenzene	ND	mg/L	0.10		100		04/05/18 00:12	106-46-7	
1,2-Dichloroethane	ND	mg/L	0.10	.5	100		04/05/18 00:12	107-06-2	
1,1-Dichloroethene	ND	mg/L	0.10	.7	100		04/05/18 00:12	75-35-4	
Tetrachloroethene	ND	mg/L	0.10	.7	100		04/05/18 00:12	127-18-4	
Trichloroethene	ND	mg/L	0.10	.5	100		04/05/18 00:12	79-01-6	
Vinyl chloride	ND	mg/L	0.10	.2	100		04/05/18 00:12	75-01-4	
Surrogates									
1,2-Dichloroethane-d4 (S)	101	%	82-120		100		04/05/18 00:12	17060-07-0	
Dibromofluoromethane (S)	99	%	79-117		100		04/05/18 00:12	1868-53-7	
4-Bromofluorobenzene (S)	99	%	82-113		100		04/05/18 00:12	460-00-4	
Toluene-d8 (S)	100	%	89-109		100		04/05/18 00:12	2037-26-5	
Percent Moisture									
Analytical Method: SM 2540 G-11/3550									
Percent Moisture	98.8	%	0.10		1		04/02/18 16:20		
2540G Total Percent Solids									
Analytical Method: SM 2540G									
Total Solids	1.2	%	0.10		1		04/02/18 16:20		N2
2540G Total Volatile Solids									
Analytical Method: SM 2540G									
Total Volatile Solids	91.7	%	0.10		1		04/02/18 16:20		D6
9045 pH Soil									
Analytical Method: EPA 9045D									
pH at 25 Degrees C	7.7	Std. Units	1.0		1		04/02/18 12:02		
351.2 Total Kjeldahl Nitrogen									
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2									
Nitrogen, Kjeldahl, Total	116000	mg/kg	41500		5	04/06/18 12:45	04/09/18 16:12	7727-37-9	
4500 Ammonia Soil, Distilled									
Analytical Method: SM 4500-NH3 G-11 Preparation Method: SM 4500-NH3 B-11									
Nitrogen, Ammonia	84600	mg/kg	1540		5	04/05/18 14:59	04/12/18 15:36	7664-41-7	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 4610111

Sample: Holding Tank 4 **Lab ID: 4610111001** Collected: 03/28/18 13:40 Received: 03/29/18 12:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
Analytical Method: SM 4500-P E-11 Preparation Method: SM 4500-P B-11									
4500PE Total Phosphorus									
Phosphorus	33900	mg/kg	743		2	04/02/18 15:24	04/03/18 17:39	7723-14-0	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 4610111

Sample: Holding Tank 4		Lab ID: 4610111002		Collected: 03/28/18 13:40		Received: 03/29/18 12:40		Matrix: Water	
Parameters	Results	Units	Report Limit	Reg. Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260B MSV		Analytical Method: EPA 8260B							
Acrolein	<25.0	ug/L	25.0		5		04/10/18 15:15	107-02-8	L2
Acrylonitrile	<5.0	ug/L	5.0		5		04/10/18 15:15	107-13-1	
Benzene	<5.0	ug/L	5.0		5		04/10/18 15:15	71-43-2	
Bromodichloromethane	<5.0	ug/L	5.0		5		04/10/18 15:15	75-27-4	
Bromoform	<5.0	ug/L	5.0		5		04/10/18 15:15	75-25-2	
Bromomethane	<5.0	ug/L	5.0		5		04/10/18 15:15	74-83-9	
Carbon tetrachloride	<5.0	ug/L	5.0		5		04/10/18 15:15	56-23-5	
Chlorobenzene	<5.0	ug/L	5.0		5		04/10/18 15:15	108-90-7	
Chloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	75-00-3	
2-Chloroethylvinyl ether	<25.0	ug/L	25.0		5		04/10/18 15:15	110-75-8	c2
Chloroform	<5.0	ug/L	5.0		5		04/10/18 15:15	67-66-3	
Chloromethane	<5.0	ug/L	5.0		5		04/10/18 15:15	74-87-3	
Dibromochloromethane	<5.0	ug/L	5.0		5		04/10/18 15:15	124-48-1	
1,2-Dichlorobenzene	<5.0	ug/L	5.0		5		04/10/18 15:15	95-50-1	
1,3-Dichlorobenzene	<5.0	ug/L	5.0		5		04/10/18 15:15	541-73-1	
1,4-Dichlorobenzene	<5.0	ug/L	5.0		5		04/10/18 15:15	106-46-7	
1,1-Dichloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	75-34-3	
1,2-Dichloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	107-06-2	
1,1-Dichloroethene	<5.0	ug/L	5.0		5		04/10/18 15:15	75-35-4	
trans-1,2-Dichloroethene	<5.0	ug/L	5.0		5		04/10/18 15:15	156-60-5	
1,2-Dichloropropane	<5.0	ug/L	5.0		5		04/10/18 15:15	78-87-5	
cis-1,3-Dichloropropene	<5.0	ug/L	5.0		5		04/10/18 15:15	10061-01-5	
trans-1,3-Dichloropropene	<5.0	ug/L	5.0		5		04/10/18 15:15	10061-02-6	
Ethylbenzene	<5.0	ug/L	5.0		5		04/10/18 15:15	100-41-4	
Methylene Chloride	<5.0	ug/L	5.0		5		04/10/18 15:15	75-09-2	
1,1,2,2-Tetrachloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	79-34-5	
Tetrachloroethene	<5.0	ug/L	5.0		5		04/10/18 15:15	127-18-4	
Toluene	<5.0	ug/L	5.0		5		04/10/18 15:15	108-88-3	
1,1,1-Trichloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	71-55-6	
1,1,2-Trichloroethane	<5.0	ug/L	5.0		5		04/10/18 15:15	79-00-5	
Trichloroethene	<5.0	ug/L	5.0		5		04/10/18 15:15	79-01-6	
Trichlorofluoromethane	<5.0	ug/L	5.0		5		04/10/18 15:15	75-69-4	
Vinyl chloride	<5.0	ug/L	5.0		5		04/10/18 15:15	75-01-4	
Surrogates									
1,2-Dichloroethane-d4 (S)	106	%	87-122		5		04/10/18 15:15	17060-07-0	D3
Dibromofluoromethane (S)	101	%	85-118		5		04/10/18 15:15	1868-53-7	
4-Bromofluorobenzene (S)	100	%	82-110		5		04/10/18 15:15	460-00-4	
Toluene-d8 (S)	97	%	85-113		5		04/10/18 15:15	2037-26-5	

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch:	19902	Analysis Method:	EPA 9056A
QC Batch Method:	EPA 9056A	Analysis Description:	9056 IC Anions
Associated Lab Samples:	4610111001		

METHOD BLANK: 79112 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<20.1	20.1	04/10/18 06:02	
Nitrate as N	mg/kg	<1.0	1.0	04/10/18 06:02	
Sulfate	mg/kg	<10.1	10.1	04/10/18 06:02	

LABORATORY CONTROL SAMPLE: 79113

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	501	481	96	80-120	
Nitrate as N	mg/kg	30.1	30.2	100	70-130	
Sulfate	mg/kg	501	485	97	70-130	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 79452 79453

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	13700	41500	41500	55900	55700	101	101	80-120	0	20	
Nitrate as N	mg/kg	<83.3	2490	2490	2510	2500	99	99	70-130	0	20	
Sulfate	mg/kg	897	41500	41500	42400	34000	100	80	70-130	22	20	R1

SAMPLE DUPLICATE: 79454

Parameter	Units	4610111001 Result	Dup Result	RPD	Max RPD	Qualifiers
Chloride	mg/kg	13700	13800	1	20	
Nitrate as N	mg/kg	<83.3	<84.5		20	
Sulfate	mg/kg	897	886	1	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19863

Analysis Method: EPA 7470A

QC Batch Method: EPA 7470A

Analysis Description: 7470 Mercury TCLP

Associated Lab Samples: 4610111001

METHOD BLANK: 76621

Matrix: Solid

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00020	0.00020	04/10/18 15:59	

LABORATORY CONTROL SAMPLE: 78990

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	.002	0.0021	103	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78991

78992

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	<0.00020	.002	.002	0.0016	0.0016	79	78	80-120	1	20	M1

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19958

Analysis Method: EPA 7471B

QC Batch Method: EPA 7471B

Analysis Description: 7471 Mercury

Associated Lab Samples: 4610111001

METHOD BLANK: 79245

Matrix: Solid

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.047	0.047	04/11/18 11:34	

LABORATORY CONTROL SAMPLE: 79246

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	.31	0.26	84	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 79247

79248

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	<1.2	7.9	8	7.3	6.4	89	77	80-120	13	20	M1

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19198	Analysis Method: EPA 6010C
QC Batch Method: EPA 3050B	Analysis Description: 6010 MET
Associated Lab Samples: 4610111001	

METHOD BLANK: 76351 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<9.8	9.8	04/03/18 10:02	
Calcium	mg/kg	<49.0	49.0	04/03/18 10:02	
Magnesium	mg/kg	<49.0	49.0	04/03/18 10:02	
Potassium	mg/kg	<49.0	49.0	04/03/18 10:02	
Sodium	mg/kg	<49.0	49.0	04/03/18 10:02	

LABORATORY CONTROL SAMPLE: 76352

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	99.6	95.0	95	80-120	
Calcium	mg/kg	996	964	97	80-120	
Magnesium	mg/kg	996	953	96	80-120	
Potassium	mg/kg	996	942	95	80-120	
Sodium	mg/kg	996	938	94	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 76353 76354

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	3630	8140	7980	13700	12700	123	113	75-125	8	20	
Calcium	mg/kg	31900	81400	79800	110000	108000	97	95	75-125	2	20	
Magnesium	mg/kg	5940	81400	79800	80600	80100	92	93	75-125	1	20	
Potassium	mg/kg	7230	81400	79800	82000	81400	92	93	75-125	1	20	
Sodium	mg/kg	11100	81400	79800	84600	83700	90	91	75-125	1	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19511	Analysis Method: EPA 6010C
QC Batch Method: EPA 3010A	Analysis Description: 6010 MET TCLP
Associated Lab Samples: 4610111001	

METHOD BLANK: 76621 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.10	0.10	04/09/18 12:04	
Barium	mg/L	<0.010	0.010	04/09/18 12:04	
Cadmium	mg/L	<0.010	0.010	04/09/18 12:04	
Chromium	mg/L	<0.050	0.050	04/09/18 12:04	
Lead	mg/L	<0.050	0.050	04/09/18 12:04	
Selenium	mg/L	<0.10	0.10	04/09/18 12:04	
Silver	mg/L	<0.010	0.010	04/09/18 12:04	
Zinc	mg/L	<0.020	0.020	04/09/18 12:04	

LABORATORY CONTROL SAMPLE: 77430

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	2	2.0	100	80-120	
Barium	mg/L	.4	0.37	94	80-120	
Cadmium	mg/L	.4	0.37	92	80-120	
Chromium	mg/L	.4	0.37	93	80-120	
Lead	mg/L	.4	0.35	86	80-120	
Selenium	mg/L	2	2.0	101	80-120	
Silver	mg/L	.4	0.44	109	80-120	
Zinc	mg/L	.4	0.37	91	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 77431 77432

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	<0.10	2	2	2.0	1.9	99	96	75-125	3	20	
Barium	mg/L	0.041	.4	.4	0.41	0.40	91	89	75-125	2	20	
Cadmium	mg/L	<0.010	.4	.4	0.36	0.35	87	85	75-125	1	20	
Chromium	mg/L	<0.050	.4	.4	0.37	0.36	90	88	75-125	2	20	
Lead	mg/L	<0.050	.4	.4	0.33	0.34	75	78	75-125	4	20	
Selenium	mg/L	<0.10	2	2	2.0	1.9	99	94	75-125	6	20	
Silver	mg/L	<0.010	.4	.4	0.40	0.39	100	97	75-125	3	20	
Zinc	mg/L	0.47	.4	.4	0.82	0.79	87	81	75-125	3	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19197	Analysis Method: EPA 6020A
QC Batch Method: EPA 3050B	Analysis Description: 6020A MET
Associated Lab Samples: 4610111001	

METHOD BLANK: 76347 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.094	0.094	04/03/18 09:54	
Cadmium	mg/kg	<0.047	0.047	04/03/18 09:54	
Chromium	mg/kg	<0.094	0.094	04/03/18 09:54	
Copper	mg/kg	<0.094	0.094	04/03/18 09:54	
Lead	mg/kg	<0.094	0.094	04/03/18 09:54	
Molybdenum	mg/kg	<0.094	0.094	04/03/18 09:54	
Nickel	mg/kg	<0.094	0.094	04/03/18 09:54	
Selenium	mg/kg	<0.094	0.094	04/03/18 10:43	
Silver	mg/kg	<0.047	0.047	04/03/18 09:54	
Zinc	mg/kg	<0.94	0.94	04/03/18 09:54	

LABORATORY CONTROL SAMPLE: 76348

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	2	1.9	94	80-120	
Cadmium	mg/kg	2	1.8	92	80-120	
Chromium	mg/kg	2	2.0	101	80-120	
Copper	mg/kg	2	2.0	99	80-120	
Lead	mg/kg	2	1.9	94	80-120	
Molybdenum	mg/kg	2	1.9	98	80-120	
Nickel	mg/kg	2	1.9	95	80-120	
Selenium	mg/kg	2	2.0	100	80-120	
Silver	mg/kg	2	1.9	94	80-120	
Zinc	mg/kg	2	2.1	107	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 76349 76350

Parameter	Units	4610140001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	<0.097	1.9	2	1.9	2.0	99	99	75-125	3	20	
Cadmium	mg/kg	<0.049	1.9	2	1.8	1.9	94	96	75-125	4	20	
Chromium	mg/kg	0.33	1.9	2	2.3	2.5	104	108	75-125	5	20	
Copper	mg/kg	2.1	1.9	2	3.2	4.4	59	119	75-125	31	20	M1, R1
Lead	mg/kg	0.27	1.9	2	2.1	2.2	93	96	75-125	5	20	
Molybdenum	mg/kg	<0.097	1.9	2	1.8	1.8	92	92	75-125	2	20	
Nickel	mg/kg	0.31	1.9	2	2.1	2.4	93	104	75-125	12	20	
Selenium	mg/kg	<0.097	1.9	2	2.4	2.6	123	129	75-125	7	20	M1
Silver	mg/kg	<0.049	1.9	2	1.9	1.9	97	95	75-125	0	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 76349 76350												
Parameter	Units	4610140001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Zinc	mg/kg	2.3	1.9	2	3.6	4.5	65	110	75-125	22	20	M1, R1

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19532	Analysis Method: EPA 8260B
QC Batch Method: EPA 8260B	Analysis Description: 8260 MSV TCLP
Associated Lab Samples: 4610111001	

METHOD BLANK: 76888 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.10	04/04/18 15:08	
1,2-Dichloroethane	mg/L	ND	0.10	04/04/18 15:08	
1,4-Dichlorobenzene	mg/L	ND	0.10	04/04/18 15:08	
2-Butanone (MEK)	mg/L	ND	2.0	04/04/18 15:08	
Benzene	mg/L	ND	0.10	04/04/18 15:08	
Carbon tetrachloride	mg/L	ND	0.10	04/04/18 15:08	
Chlorobenzene	mg/L	ND	0.10	04/04/18 15:08	
Chloroform	mg/L	ND	0.10	04/04/18 15:08	
Tetrachloroethene	mg/L	ND	0.10	04/04/18 15:08	
Trichloroethene	mg/L	ND	0.10	04/04/18 15:08	
Vinyl chloride	mg/L	ND	0.10	04/04/18 15:08	
1,2-Dichloroethane-d4 (S)	%	102	82-120	04/04/18 15:08	
4-Bromofluorobenzene (S)	%	100	82-113	04/04/18 15:08	
Dibromofluoromethane (S)	%	99	79-117	04/04/18 15:08	
Toluene-d8 (S)	%	101	89-109	04/04/18 15:08	

LABORATORY CONTROL SAMPLE: 77519

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	4	4.2	105	78-124	
1,2-Dichloroethane	mg/L	4	4.0	101	78-125	
1,4-Dichlorobenzene	mg/L	4	3.9	98	84-116	
2-Butanone (MEK)	mg/L	4	4.2	106	58-139	
Benzene	mg/L	4	4.2	104	77-122	
Carbon tetrachloride	mg/L	4	4.2	106	77-131	
Chlorobenzene	mg/L	4	4.0	101	85-115	
Chloroform	mg/L	4	4.3	107	78-127	
Tetrachloroethene	mg/L	4	4.1	101	82-119	
Trichloroethene	mg/L	4	4.1	103	81-122	
Vinyl chloride	mg/L	4	4.0	100	76-125	
1,2-Dichloroethane-d4 (S)	%			102	82-120	
4-Bromofluorobenzene (S)	%			102	82-113	
Dibromofluoromethane (S)	%			100	79-117	
Toluene-d8 (S)	%			100	89-109	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 77520 77521											
Parameter	Units	469827001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
1,1-Dichloroethene	mg/L	ND	4	4	3.8	3.9	95	97	58-142	2	20
1,2-Dichloroethane	mg/L	ND	4	4	3.9	3.9	98	96	75-127	2	30
1,4-Dichlorobenzene	mg/L	ND	4	4	3.6	3.6	90	91	82-116	1	40
2-Butanone (MEK)	mg/L	ND	4	4	3.9	4.0	99	100	12-189	2	20
Benzene	mg/L	ND	4	4	3.9	4.0	98	100	83-122	1	18
Carbon tetrachloride	mg/L	ND	4	4	4.0	4.2	101	104	70-133	2	18
Chlorobenzene	mg/L	ND	4	4	3.8	3.8	94	95	82-117	1	27
Chloroform	mg/L	ND	4	4	3.9	4.0	99	101	80-123	2	17
Tetrachloroethene	mg/L	ND	4	4	3.7	3.8	92	94	77-122	2	31
Trichloroethene	mg/L	ND	4	4	3.9	3.9	96	98	67-138	2	21
Vinyl chloride	mg/L	ND	4	4	3.8	4.0	96	100	70-130	5	25
1,2-Dichloroethane-d4 (S)	%						102	100	82-120		
4-Bromofluorobenzene (S)	%						103	103	82-113		
Dibromofluoromethane (S)	%						102	100	79-117		
Toluene-d8 (S)	%						103	103	89-109		

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19961

Analysis Method: EPA 8260B

QC Batch Method: EPA 8260B

Analysis Description: 8260B MSV

Associated Lab Samples: 4610111002

METHOD BLANK: 79257

Matrix: Water

Associated Lab Samples: 4610111002

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1-Trichloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
1,1,2,2-Tetrachloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
1,1,2-Trichloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
1,1-Dichloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
1,1-Dichloroethene	ug/L	<1.0	1.0	04/10/18 09:07	
1,2-Dichlorobenzene	ug/L	<1.0	1.0	04/10/18 09:07	
1,2-Dichloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
1,2-Dichloropropane	ug/L	<1.0	1.0	04/10/18 09:07	
1,3-Dichlorobenzene	ug/L	<1.0	1.0	04/10/18 09:07	
1,4-Dichlorobenzene	ug/L	<1.0	1.0	04/10/18 09:07	
2-Chloroethylvinyl ether	ug/L	<5.0	5.0	04/10/18 09:07	
Acrolein	ug/L	<5.0	5.0	04/10/18 09:07	
Acrylonitrile	ug/L	<1.0	1.0	04/10/18 09:07	
Benzene	ug/L	<1.0	1.0	04/10/18 09:07	
Bromodichloromethane	ug/L	<1.0	1.0	04/10/18 09:07	
Bromoform	ug/L	<1.0	1.0	04/10/18 09:07	
Bromomethane	ug/L	<1.0	1.0	04/10/18 09:07	
Carbon tetrachloride	ug/L	<1.0	1.0	04/10/18 09:07	
Chlorobenzene	ug/L	<1.0	1.0	04/10/18 09:07	
Chloroethane	ug/L	<1.0	1.0	04/10/18 09:07	
Chloroform	ug/L	<1.0	1.0	04/10/18 09:07	
Chloromethane	ug/L	<1.0	1.0	04/10/18 09:07	
cis-1,3-Dichloropropene	ug/L	<1.0	1.0	04/10/18 09:07	
Dibromochloromethane	ug/L	<1.0	1.0	04/10/18 09:07	
Ethylbenzene	ug/L	<1.0	1.0	04/10/18 09:07	
Methylene Chloride	ug/L	<1.0	1.0	04/10/18 09:07	
Tetrachloroethene	ug/L	<1.0	1.0	04/10/18 09:07	
Toluene	ug/L	<1.0	1.0	04/10/18 09:07	
trans-1,2-Dichloroethene	ug/L	<1.0	1.0	04/10/18 09:07	
trans-1,3-Dichloropropene	ug/L	<1.0	1.0	04/10/18 09:07	
Trichloroethene	ug/L	<1.0	1.0	04/10/18 09:07	
Trichlorofluoromethane	ug/L	<1.0	1.0	04/10/18 09:07	
Vinyl chloride	ug/L	<1.0	1.0	04/10/18 09:07	
1,2-Dichloroethane-d4 (S)	%	105	87-122	04/10/18 09:07	
4-Bromofluorobenzene (S)	%	100	82-110	04/10/18 09:07	
Dibromofluoromethane (S)	%	102	85-118	04/10/18 09:07	
Toluene-d8 (S)	%	97	85-113	04/10/18 09:07	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

LABORATORY CONTROL SAMPLE: 79258

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1-Trichloroethane	ug/L	40	43.5	109	81-122	
1,1,2,2-Tetrachloroethane	ug/L	40	43.6	109	70-137	
1,1,2-Trichloroethane	ug/L	40	42.6	107	83-121	
1,1-Dichloroethane	ug/L	40	42.4	106	80-118	
1,1-Dichloroethene	ug/L	40	43.1	108	77-123	
1,2-Dichlorobenzene	ug/L	40	42.1	105	81-124	
1,2-Dichloroethane	ug/L	40	43.1	108	81-122	
1,2-Dichloropropane	ug/L	40	43.0	108	82-122	
1,3-Dichlorobenzene	ug/L	40	42.0	105	81-124	
1,4-Dichlorobenzene	ug/L	40	41.8	104	79-122	
2-Chloroethylvinyl ether	ug/L	40	51.2	128	65-130	
Acrolein	ug/L	40	14.7	37	48-146	L2
Acrylonitrile	ug/L	40	43.8	109	73-129	
Benzene	ug/L	40	40.8	102	84-119	
Bromodichloromethane	ug/L	40	38.2	96	82-124	
Bromoform	ug/L	40	41.3	103	65-123	
Bromomethane	ug/L	40	46.6	117	55-142	
Carbon tetrachloride	ug/L	40	39.1	98	79-127	
Chlorobenzene	ug/L	40	42.1	105	84-118	
Chloroethane	ug/L	40	39.7	99	76-124	
Chloroform	ug/L	40	42.7	107	82-119	
Chloromethane	ug/L	40	41.3	103	73-125	
cis-1,3-Dichloropropene	ug/L	40	38.3	96	77-120	
Dibromochloromethane	ug/L	40	38.1	95	74-121	
Ethylbenzene	ug/L	40	42.3	106	87-119	
Methylene Chloride	ug/L	40	42.6	106	75-129	
Tetrachloroethene	ug/L	40	41.6	104	81-117	
Toluene	ug/L	40	41.0	103	85-118	
trans-1,2-Dichloroethene	ug/L	40	41.7	104	76-126	
trans-1,3-Dichloropropene	ug/L	40	37.4	94	73-116	
Trichloroethene	ug/L	40	42.9	107	82-119	
Trichlorofluoromethane	ug/L	40	40.4	101	76-128	
Vinyl chloride	ug/L	40	41.1	103	77-123	
1,2-Dichloroethane-d4 (S)	%			102	87-122	
4-Bromofluorobenzene (S)	%			102	82-110	
Dibromofluoromethane (S)	%			103	85-118	
Toluene-d8 (S)	%			97	85-113	

LABORATORY CONTROL SAMPLE: 79812

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,2-Dichloroethane-d4 (S)	%			103	87-122	
4-Bromofluorobenzene (S)	%			100	82-110	
Dibromofluoromethane (S)	%			100	85-118	
Toluene-d8 (S)	%			96	85-113	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19824	Analysis Method: EPA 8081B
QC Batch Method: EPA 3545A	Analysis Description: 8081 GCS Pesticides
Associated Lab Samples: 4610111001	

METHOD BLANK: 78863 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<16.9	16.9	04/18/18 05:01	
4,4'-DDE	ug/kg	<16.9	16.9	04/18/18 05:01	
4,4'-DDT	ug/kg	<16.9	16.9	04/18/18 05:01	
Aldrin	ug/kg	<16.9	16.9	04/18/18 05:01	
alpha-BHC	ug/kg	<16.9	16.9	04/18/18 05:01	
alpha-Chlordane	ug/kg	<16.9	16.9	04/18/18 05:01	
beta-BHC	ug/kg	<16.9	16.9	04/18/18 05:01	
Chlordane (Technical)	ug/kg	<49.6	49.6	04/18/18 05:01	
delta-BHC	ug/kg	<16.9	16.9	04/18/18 05:01	
Dieldrin	ug/kg	<16.9	16.9	04/18/18 05:01	
Endosulfan I	ug/kg	<16.9	16.9	04/18/18 05:01	
Endosulfan II	ug/kg	<16.9	16.9	04/18/18 05:01	
Endosulfan sulfate	ug/kg	<16.9	16.9	04/18/18 05:01	
Endrin	ug/kg	<16.9	16.9	04/18/18 05:01	
Endrin aldehyde	ug/kg	<16.9	16.9	04/18/18 05:01	
gamma-BHC (Lindane)	ug/kg	<16.9	16.9	04/18/18 05:01	
gamma-Chlordane	ug/kg	<16.9	16.9	04/18/18 05:01	
Heptachlor	ug/kg	<16.9	16.9	04/18/18 05:01	
Heptachlor epoxide	ug/kg	<16.9	16.9	04/18/18 05:01	
Toxaphene	ug/kg	<169	169	04/18/18 05:01	
Decachlorobiphenyl (S)	%	73	58-128	04/18/18 05:01	
Tetrachloro-m-xylene (S)	%	65	63-125	04/18/18 05:01	

LABORATORY CONTROL SAMPLE: 78864

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	26.5	23.1	87	66-138	
4,4'-DDE	ug/kg	26.5	23.1	87	68-136	
4,4'-DDT	ug/kg	26.5	27.6	104	63-131	
Aldrin	ug/kg	26.5	21.9	83	69-127	
alpha-BHC	ug/kg	26.5	21.2	80	68-131	
alpha-Chlordane	ug/kg	26.5	21.6	82	68-127	
beta-BHC	ug/kg	26.5	20.5	77	65-130	
delta-BHC	ug/kg	26.5	21.9	83	68-132	
Dieldrin	ug/kg	26.5	25.1	95	70-132	
Endosulfan I	ug/kg	26.5	22.5	85	45-135	
Endosulfan II	ug/kg	26.5	23.3	88	59-113	
Endosulfan sulfate	ug/kg	26.5	25.4	96	62-126	
Endrin	ug/kg	26.5	25.2	95	51-134	
Endrin aldehyde	ug/kg	26.5	24.0	90	55-128	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

LABORATORY CONTROL SAMPLE: 78864

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	26.5	23.7	89	68-131	
gamma-Chlordane	ug/kg	26.5	22.0	83	77-113	
Heptachlor	ug/kg	26.5	23.4	88	70-129	
Heptachlor epoxide	ug/kg	26.5	24.3	92	68-130	
Decachlorobiphenyl (S)	%.			80	58-128	
Tetrachloro-m-xylene (S)	%.			67	63-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78865 78866

Parameter	Units	4610390005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	464	462	<295	<295	45	48	69-140		33	M1
4,4'-DDE	ug/kg	ND	464	462	<295	<295	53	56	70-136		25	M1
4,4'-DDT	ug/kg	ND	464	462	<295	<295	51	59	61-143		36	M1
Aldrin	ug/kg	ND	464	462	1010	1110	189	211	64-135	9	22	M1
alpha-BHC	ug/kg	ND	464	462	<295	<295	49	55	67-133		23	M1
alpha-Chlordane	ug/kg	ND	464	462	<295	<295	57	60	69-130		25	M1
beta-BHC	ug/kg	ND	464	462	368	365	66	65	65-137	1	30	
delta-BHC	ug/kg	ND	464	462	<295	<295	52	60	73-135		20	M1
Dieldrin	ug/kg	ND	464	462	<295	<295	42	46	67-130		28	M1
Endosulfan I	ug/kg	ND	464	462	<295	<295	45	50	44-130		28	
Endosulfan II	ug/kg	ND	464	462	<295	<295	41	47	48-135		32	M1
Endosulfan sulfate	ug/kg	ND	464	462	<295	337	63	73	66-141		23	M1
Endrin	ug/kg	ND	464	462	<295	<295	52	59	64-145		26	M1
Endrin aldehyde	ug/kg	ND	464	462	<295	<295	35	37	59-130		35	M1
gamma-BHC (Lindane)	ug/kg	ND	464	462	<295	300	50	54	66-136		21	M1
gamma-Chlordane	ug/kg	0.76 mg/kg	464	462	829	984	14	48	67-130	17	24	M1
Heptachlor	ug/kg	ND	464	462	<295	<295	53	60	56-141		35	M1
Heptachlor epoxide	ug/kg	ND	464	462	<295	<295	55	61	69-130		21	M1
Decachlorobiphenyl (S)	%.						44	45	58-128			S0
Tetrachloro-m-xylene (S)	%.						42	45	63-125			S0

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19712	Analysis Method: EPA 8081B
QC Batch Method: EPA 3510C	Analysis Description: 8081B GCS TCLP Pesticides
Associated Lab Samples: 4610111001	

METHOD BLANK: 76622 Matrix: Water
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	ND	0.025	04/18/18 01:59	
Endrin	mg/L	ND	0.0050	04/18/18 01:59	
gamma-BHC (Lindane)	mg/L	ND	0.0050	04/18/18 01:59	
Heptachlor	mg/L	ND	0.0050	04/18/18 01:59	
Heptachlor epoxide	mg/L	ND	0.0050	04/18/18 01:59	
Methoxychlor	mg/L	ND	0.0050	04/18/18 01:59	CU
Toxaphene	mg/L	ND	0.10	04/18/18 01:59	
Decachlorobiphenyl (S)	%	85	61-115	04/18/18 01:59	
Tetrachloro-m-xylene (S)	%	76	68-111	04/18/18 01:59	

LABORATORY CONTROL SAMPLE: 78387

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	.04	0.045	112	67-152	
gamma-BHC (Lindane)	mg/L	.04	0.037	93	74-138	
Heptachlor	mg/L	.04	0.042	104	71-139	
Heptachlor epoxide	mg/L	.04	0.043	107	75-139	
Methoxychlor	mg/L	.04	0.051	128	65-155	
Decachlorobiphenyl (S)	%			86	61-115	
Tetrachloro-m-xylene (S)	%			76	68-111	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78388 78389

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
Endrin	mg/L	ND	.04	.04	0.046	0.042	114	105	49-155	8	30
gamma-BHC (Lindane)	mg/L	ND	.04	.04	0.041	0.035	102	87	60-141	16	30
Heptachlor	mg/L	ND	.04	.04	0.041	0.041	101	101	55-144	0	30
Heptachlor epoxide	mg/L	ND	.04	.04	0.043	0.041	108	101	61-140	7	30
Methoxychlor	mg/L	ND	.04	.04	0.050	0.047	125	116	49-159	7	30
Decachlorobiphenyl (S)	%						83	80	61-115		
Tetrachloro-m-xylene (S)	%						76	74	68-111		

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19573	Analysis Method: EPA 8082A
QC Batch Method: EPA 3545A	Analysis Description: 8082A GCS PCB
Associated Lab Samples: 4610111001	

METHOD BLANK: 77786 Matrix: Solid
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1221 (Aroclor 1221)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1232 (Aroclor 1232)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1242 (Aroclor 1242)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1248 (Aroclor 1248)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1254 (Aroclor 1254)	ug/kg	<33.0	33.0	04/06/18 13:01	
PCB-1260 (Aroclor 1260)	ug/kg	<33.0	33.0	04/06/18 13:01	
Decachlorobiphenyl (S)	%.	92	45-135	04/06/18 13:01	
Tetrachloro-m-xylene (S)	%.	90	56-123	04/06/18 13:01	

LABORATORY CONTROL SAMPLE: 77787

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	200	179	89	68-129	
PCB-1260 (Aroclor 1260)	ug/kg	200	175	88	60-140	
Decachlorobiphenyl (S)	%.			80	45-135	
Tetrachloro-m-xylene (S)	%.			88	56-123	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 77788 77789

Parameter	Units	469837001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	218	218	179	186	82	85	49-128	4	30	
PCB-1260 (Aroclor 1260)	ug/kg	ND	218	218	171	172	79	79	48-138	1	30	
Decachlorobiphenyl (S)	%.						66	61	45-135			
Tetrachloro-m-xylene (S)	%.						73	76	56-123			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch:	19826	Analysis Method:	EPA 8151A
QC Batch Method:	EPA 8151A	Analysis Description:	8151 GCS TCLP Herbicides
Associated Lab Samples:	4610111001		

METHOD BLANK: 76622 Matrix: Water

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	ND	0.062	04/11/18 08:18	
2,4-D	mg/L	ND	0.62	04/11/18 08:18	
2,4-DCAA (S)	%.	95	33-153	04/11/18 08:18	

LABORATORY CONTROL SAMPLE: 78868

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	.5	0.47	95	50-152	
2,4-D	mg/L	5	4.5	90	66-148	
2,4-DCAA (S)	%.			100	33-153	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78869 78870

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4,5-TP (Silvex)	mg/L	ND	.5	.5	0.44	0.49	87	99	25-152	12	20	
2,4-D	mg/L	ND	5	5	3.9	4.4	78	88	29-173	12	20	
2,4-DCAA (S)	%.						89	98	33-153			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19230	Analysis Method: EPA 8270C
QC Batch Method: EPA 3550C	Analysis Description: 8270C Solid MSSV
Associated Lab Samples: 4610111001	

METHOD BLANK: 76542	Matrix: Solid
Associated Lab Samples: 4610111001	

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,2,4-Trichlorobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
1,2-Dichlorobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
1,2-Diphenylhydrazine	ug/kg	<17.0	17.0	04/13/18 14:03	
1,3-Dichlorobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
1,4-Dichlorobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
2,4,6-Trichlorophenol	ug/kg	<17.0	17.0	04/13/18 14:03	
2,4-Dichlorophenol	ug/kg	<33.0	33.0	04/13/18 14:03	
2,4-Dimethylphenol	ug/kg	<170	170	04/13/18 14:03	
2,4-Dinitrophenol	ug/kg	<170	170	04/13/18 14:03	
2,4-Dinitrotoluene	ug/kg	<33.0	33.0	04/13/18 14:03	
2,6-Dinitrotoluene	ug/kg	<17.0	17.0	04/13/18 14:03	
2-Chloroaniline	ug/kg	<17.0	17.0	04/13/18 14:03	
2-Chloronaphthalene	ug/kg	<17.0	17.0	04/13/18 14:03	
2-Chlorophenol	ug/kg	<17.0	17.0	04/13/18 14:03	
2-Nitrophenol	ug/kg	<17.0	17.0	04/13/18 14:03	
3,3'-Dichlorobenzidine	ug/kg	<670	670	04/13/18 14:03	
4,6-Dinitro-2-methylphenol	ug/kg	<170	170	04/13/18 14:03	
4-Bromophenylphenyl ether	ug/kg	<17.0	17.0	04/13/18 14:03	
4-Chloro-3-methylphenol	ug/kg	<17.0	17.0	04/13/18 14:03	
4-Chloroaniline	ug/kg	<67.0	67.0	04/13/18 14:03	
4-Chlorophenylphenyl ether	ug/kg	<17.0	17.0	04/13/18 14:03	
4-Nitrophenol	ug/kg	<670	670	04/13/18 14:03	
Acenaphthene	ug/kg	<17.0	17.0	04/13/18 14:03	
Acenaphthylene	ug/kg	<17.0	17.0	04/13/18 14:03	
Anthracene	ug/kg	<17.0	17.0	04/13/18 14:03	
Benzidine	ug/kg	<670	670	04/13/18 14:03	
Benzo(a)anthracene	ug/kg	<17.0	17.0	04/13/18 14:03	
Benzo(a)pyrene	ug/kg	<17.0	17.0	04/13/18 14:03	
Benzo(b)fluoranthene	ug/kg	<17.0	17.0	04/13/18 14:03	
Benzo(g,h,i)perylene	ug/kg	<33.0	33.0	04/13/18 14:03	
Benzo(k)fluoranthene	ug/kg	<17.0	17.0	04/13/18 14:03	
bis(2-Chloroethoxy)methane	ug/kg	<17.0	17.0	04/13/18 14:03	
bis(2-Chloroethyl) ether	ug/kg	<17.0	17.0	04/13/18 14:03	
bis(2-Chloroisopropyl) ether	ug/kg	<17.0	17.0	04/13/18 14:03	
bis(2-Ethylhexyl)phthalate	ug/kg	<33.0	33.0	04/13/18 14:03	
Butylbenzylphthalate	ug/kg	<33.0	33.0	04/13/18 14:03	
Chrysene	ug/kg	<17.0	17.0	04/13/18 14:03	
Di-n-butylphthalate	ug/kg	<67.0	67.0	04/13/18 14:03	
Di-n-octylphthalate	ug/kg	<17.0	17.0	04/13/18 14:03	
Dibenz(a,h)anthracene	ug/kg	<33.0	33.0	04/13/18 14:03	
Dibenzofuran	ug/kg	<17.0	17.0	04/13/18 14:03	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

METHOD BLANK: 76542
Associated Lab Samples: 4610111001

Matrix: Solid

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Diethylphthalate	ug/kg	<17.0	17.0	04/13/18 14:03	
Dimethylphthalate	ug/kg	<17.0	17.0	04/13/18 14:03	
Fluoranthene	ug/kg	<17.0	17.0	04/13/18 14:03	
Fluorene	ug/kg	<33.0	33.0	04/13/18 14:03	
Hexachloro-1,3-butadiene	ug/kg	<17.0	17.0	04/13/18 14:03	
Hexachlorobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
Hexachlorocyclopentadiene	ug/kg	<17.0	17.0	04/13/18 14:03	
Hexachloroethane	ug/kg	<17.0	17.0	04/13/18 14:03	
Indeno(1,2,3-cd)pyrene	ug/kg	<33.0	33.0	04/13/18 14:03	
Isophorone	ug/kg	<17.0	17.0	04/13/18 14:03	
N-Nitroso-di-n-propylamine	ug/kg	<17.0	17.0	04/13/18 14:03	
N-Nitrosodimethylamine	ug/kg	<33.0	33.0	04/13/18 14:03	
N-Nitrosodiphenylamine	ug/kg	<17.0	17.0	04/13/18 14:03	
Naphthalene	ug/kg	<17.0	17.0	04/13/18 14:03	
Nitrobenzene	ug/kg	<17.0	17.0	04/13/18 14:03	
Pentachlorophenol	ug/kg	<33.0	33.0	04/13/18 14:32	
Phenanthrene	ug/kg	<17.0	17.0	04/13/18 14:03	
Phenol	ug/kg	<170	170	04/13/18 14:03	
Pyrene	ug/kg	<17.0	17.0	04/13/18 14:03	
2,4,6-Tribromophenol (S)	%	71	12-124	04/13/18 14:03	
2-Fluorobiphenyl (S)	%	75	46-122	04/13/18 14:03	
2-Fluorophenol (S)	%	68	33-113	04/13/18 14:03	
Nitrobenzene-d5 (S)	%	68	33-131	04/13/18 14:03	
o-Terphenyl (S)	%	80	20-155	04/13/18 14:03	
Phenol-d6 (S)	%	28	30-115	04/13/18 14:03	S0

LABORATORY CONTROL SAMPLE: 76543

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,2,4-Trichlorobenzene	ug/kg	333	275	82	51-110	
1,2-Dichlorobenzene	ug/kg	333	264	79	63-115	
1,2-Diphenylhydrazine	ug/kg	333	276	83	68-125	
1,3-Dichlorobenzene	ug/kg	333	260	78	54-113	
1,4-Dichlorobenzene	ug/kg	333	255	76	61-111	
2,4,6-Trichlorophenol	ug/kg	333	241	72	45-128	
2,4-Dichlorophenol	ug/kg	333	168	50	50-128	
2,4-Dimethylphenol	ug/kg	333	<170	46	40-122	
2,4-Dinitrophenol	ug/kg	333	320	96	25-105	
2,4-Dinitrotoluene	ug/kg	333	285	86	51-128	
2,6-Dinitrotoluene	ug/kg	333	285	86	61-119	
2-Chloroaniline	ug/kg	333	40.0	12	50-150	L2
2-Chloronaphthalene	ug/kg	333	273	82	67-111	
2-Chlorophenol	ug/kg	333	257	77	62-118	
2-Nitrophenol	ug/kg	333	292	88	55-115	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

LABORATORY CONTROL SAMPLE: 76543

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
3,3'-Dichlorobenzidine	ug/kg	667	<670	44	10-107	
4,6-Dinitro-2-methylphenol	ug/kg	333	326	98	26-136	
4-Bromophenylphenyl ether	ug/kg	333	300	90	61-124	
4-Chloro-3-methylphenol	ug/kg	333	242	73	57-124	
4-Chloroaniline	ug/kg	333	<67.0	15	7-75	
4-Chlorophenylphenyl ether	ug/kg	333	266	80	62-114	
4-Nitrophenol	ug/kg	333	<670	91	36-131	
Acenaphthene	ug/kg	333	280	84	55-113	
Acenaphthylene	ug/kg	333	296	89	56-138	
Anthracene	ug/kg	333	287	86	63-134	
Benzidine	ug/kg	667	<670	5	10-74	L2
Benzo(a)anthracene	ug/kg	333	320	96	53-142	
Benzo(a)pyrene	ug/kg	333	295	88	54-136	
Benzo(b)fluoranthene	ug/kg	333	288	86	49-146	
Benzo(g,h,i)perylene	ug/kg	333	326	98	47-141	
Benzo(k)fluoranthene	ug/kg	333	275	83	56-136	
bis(2-Chloroethoxy)methane	ug/kg	333	257	77	57-121	
bis(2-Chloroethyl) ether	ug/kg	333	261	78	54-112	
bis(2-Chloroisopropyl) ether	ug/kg	333	256	77	62-116	
bis(2-Ethylhexyl)phthalate	ug/kg	333	342	103	50-140	
Butylbenzylphthalate	ug/kg	333	343	103	51-145	
Chrysene	ug/kg	333	300	90	66-137	
Di-n-butylphthalate	ug/kg	333	357	107	65-140	
Di-n-octylphthalate	ug/kg	333	342	102	63-132	
Dibenz(a,h)anthracene	ug/kg	333	322	97	52-142	
Dibenzofuran	ug/kg	333	275	83	65-119	
Diethylphthalate	ug/kg	333	273	82	59-128	
Dimethylphthalate	ug/kg	333	288	86	66-122	
Fluoranthene	ug/kg	333	303	91	66-140	
Fluorene	ug/kg	333	280	84	60-131	
Hexachloro-1,3-butadiene	ug/kg	333	283	85	56-128	
Hexachlorobenzene	ug/kg	333	311	93	34-141	
Hexachlorocyclopentadiene	ug/kg	333	230	69	34-124	
Hexachloroethane	ug/kg	333	257	77	60-111	
Indeno(1,2,3-cd)pyrene	ug/kg	333	334	100	53-135	
Isophorone	ug/kg	333	255	76	55-127	
N-Nitroso-di-n-propylamine	ug/kg	333	294	88	48-127	
N-Nitrosodimethylamine	ug/kg	333	206	62	27-152	
N-Nitrosodiphenylamine	ug/kg	333	283	85	33-109	
Naphthalene	ug/kg	333	291	87	52-128	
Nitrobenzene	ug/kg	333	273	82	56-109	
Pentachlorophenol	ug/kg	333	328	98	19-117	
Phenanthrene	ug/kg	333	275	82	58-134	
Phenol	ug/kg	333	<170	34	53-120	L2
Pyrene	ug/kg	333	294	88	60-132	
2,4,6-Tribromophenol (S)	%			94	12-124	
2-Fluorobiphenyl (S)	%			85	46-122	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

LABORATORY CONTROL SAMPLE: 76543

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2-Fluorophenol (S)	%.			83	33-113	
Nitrobenzene-d5 (S)	%.			86	33-131	
o-Terphenyl (S)	%.			87	20-155	
Phenol-d6 (S)	%.			37	30-115	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 4610111

QC Batch: 19713	Analysis Method: EPA 8270C
QC Batch Method: EPA 3510C	Analysis Description: 8270C TCLP MSSV
Associated Lab Samples: 4610111001	

METHOD BLANK: 76622 Matrix: Water
Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-Trichlorophenol	mg/L	ND	0.60	04/14/18 01:23	
2,4,6-Trichlorophenol	mg/L	ND	0.20	04/14/18 01:23	
2,4-Dinitrotoluene	mg/L	ND	0.10	04/14/18 01:23	
2-Methylphenol(o-Cresol)	mg/L	ND	0.20	04/14/18 01:23	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	0.40	04/14/18 01:23	
Hexachloro-1,3-butadiene	mg/L	ND	0.10	04/14/18 01:23	
Hexachlorobenzene	mg/L	ND	0.10	04/14/18 01:23	
Hexachloroethane	mg/L	ND	0.20	04/14/18 01:23	
Nitrobenzene	mg/L	ND	0.20	04/14/18 01:23	
Pentachlorophenol	mg/L	ND	1.5	04/14/18 01:23	
Pyridine	mg/L	ND	1.0	04/14/18 01:23	
2,4,6-Tribromophenol (S)	%	97	20-142	04/14/18 01:23	
2-Fluorobiphenyl (S)	%	96	41-132	04/14/18 01:23	
2-Fluorophenol (S)	%	88	10-123	04/14/18 01:23	
Nitrobenzene-d5 (S)	%	100	38-141	04/14/18 01:23	
o-Terphenyl (S)	%	104	39-148	04/14/18 01:23	
Phenol-d6 (S)	%	79	10-112	04/14/18 01:23	

LABORATORY CONTROL SAMPLE: 78390

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-Trichlorophenol	mg/L	2	1.8	89	36-137	
2,4,6-Trichlorophenol	mg/L	2	1.8	90	41-135	
2,4-Dinitrotoluene	mg/L	2	1.7	84	42-125	
2-Methylphenol(o-Cresol)	mg/L	2	1.8	92	37-125	
3&4-Methylphenol(m&p Cresol)	mg/L	4	3.7	92	45-125	
Hexachloro-1,3-butadiene	mg/L	2	1.8	92	39-120	
Hexachlorobenzene	mg/L	2	1.9	94	26-141	
Hexachloroethane	mg/L	2	1.7	85	34-125	
Nitrobenzene	mg/L	2	2.0	99	52-129	
Pentachlorophenol	mg/L	2	1.3J	65	28-108	
Pyridine	mg/L	2	1.6	79	5-120	
2,4,6-Tribromophenol (S)	%			86	20-142	
2-Fluorobiphenyl (S)	%			95	41-132	
2-Fluorophenol (S)	%			88	10-123	
Nitrobenzene-d5 (S)	%			103	38-141	
o-Terphenyl (S)	%			100	39-148	
Phenol-d6 (S)	%			80	10-112	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78391											
78392											
Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
2,4,5-Trichlorophenol	mg/L	ND	2	2	1.8	1.7	90	86	35-132	5	25
2,4,6-Trichlorophenol	mg/L	ND	2	2	1.7	1.8	85	88	31-133	3	25
2,4-Dinitrotoluene	mg/L	ND	2	2	1.6	1.7	81	84	41-130	4	16
2-Methylphenol(o-Cresol)	mg/L	ND	2	2	1.8	1.8	89	91	38-118	2	20
3&4-Methylphenol(m&p Cresol)	mg/L	ND	4	4	3.5	3.7	88	93	26-89	5	37 M1
Hexachloro-1,3-butadiene	mg/L	ND	2	2	1.7	1.8	85	90	33-107	5	20
Hexachlorobenzene	mg/L	ND	2	2	1.8	1.8	90	91	45-135	1	17
Hexachloroethane	mg/L	ND	2	2	1.7	1.7	84	86	46-134	2	19
Nitrobenzene	mg/L	ND	2	2	2.0	1.9	99	97	48-135	1	16
Pentachlorophenol	mg/L	ND	2	2	.93J	.94J	46	47	26-111		29
Pyridine	mg/L	ND	2	2	1.2	1.1	62	56	27-96	10	20
2,4,6-Tribromophenol (S)	%.						87	92	20-142		
2-Fluorobiphenyl (S)	%.						95	99	41-132		
2-Fluorophenol (S)	%.						80	87	10-123		
Nitrobenzene-d5 (S)	%.						100	104	38-141		
o-Terphenyl (S)	%.						93	99	39-148		
Phenol-d6 (S)	%.						81	82	10-112		

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19256

Analysis Method: SM 2540 G-11/3550

QC Batch Method: SM 2540 G-11/3550

Analysis Description: Dry Weight/Percent Moisture

Associated Lab Samples: 4610111001

SAMPLE DUPLICATE: 76616

Parameter	Units	50193223001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	11.2	9.4	17	20	

SAMPLE DUPLICATE: 76617

Parameter	Units	50193223008 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	11.6	10.3	12	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19254

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Solids

Associated Lab Samples: 4610111001

SAMPLE DUPLICATE: 76614

Parameter	Units	4610111001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	1.2	1.2	4	5	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19255

Analysis Method: SM 2540G

QC Batch Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Associated Lab Samples: 4610111001

SAMPLE DUPLICATE: 76615

Parameter	Units	4610111001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	91.7	99.1	8	5	D6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch:	19277	Analysis Method:	EPA 9045D
QC Batch Method:	EPA 9045D	Analysis Description:	9045 pH
Associated Lab Samples:	4610111001		

LABORATORY CONTROL SAMPLE: 76658

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
pH at 25 Degrees C	Std. Units	6	6.0	100	99-101	

SAMPLE DUPLICATE: 76659

Parameter	Units	4610111001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.7	7.7	0	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19745

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Associated Lab Samples: 4610111001

METHOD BLANK: 78487

Matrix: Solid

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<100	100	04/09/18 14:52	

LABORATORY CONTROL SAMPLE: 78488

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	250	234	93	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78489

78490

Parameter	Units	4610111001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	116000	20800	20800	138000	138000	103	107	90-110	0	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19636

Analysis Method: SM 4500-NH3 G-11

QC Batch Method: SM 4500-NH3 B-11

Analysis Description: 4500 Ammonia, Distilled

Associated Lab Samples: 4610111001

METHOD BLANK: 78015

Matrix: Solid

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<0.99	0.99	04/12/18 16:20	

LABORATORY CONTROL SAMPLE: 78016

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.7	18.9	96	82-112	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 78017

78018

Parameter	Units	4610270001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	1490	730	733	1770	2810	38	179	49-148	45	20	M1, R1

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 4610111

QC Batch: 19313

Analysis Method: SM 4500-P E-11

QC Batch Method: SM 4500-P B-11

Analysis Description: 4500PE Soil Phosphorus

Associated Lab Samples: 4610111001

METHOD BLANK: 76772

Matrix: Solid

Associated Lab Samples: 4610111001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Phosphorus	mg/kg	<5.0	5.0	04/03/18 16:34	

LABORATORY CONTROL SAMPLE: 76773

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phosphorus	mg/kg	100	97.2	97	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 76774

76775

Parameter	Units	469457001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phosphorus	mg/kg	48800	2000	1940	38500	46100	-515	-137	90-110	18	20	M6

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QUALIFIERS

Project: Biosolids
Pace Project No.: 4610111

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
 ND - Not Detected at or above adjusted reporting limit.
 TNTC - Too Numerous To Count
 J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
 MDL - Adjusted Method Detection Limit.
 PQL - Practical Quantitation Limit.
 RL - Reporting Limit.
 S - Surrogate
 1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
 Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
 LCS(D) - Laboratory Control Sample (Duplicate)
 MS(D) - Matrix Spike (Duplicate)
 DUP - Sample Duplicate
 RPD - Relative Percent Difference
 NC - Not Calculable.
 SG - Silica Gel - Clean-Up
 U - Indicates the compound was analyzed for, but not detected.
 N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
 Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
 TNI - The NELAC Institute.

ANALYTE QUALIFIERS

CU	The continuing calibration for this analyte is above laboratory acceptance limits. Analyte was not detected above the reporting limit in any of the associated samples.
D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
D6	The precision between the sample and sample duplicate exceeded laboratory control limits.
L2	Analyte recovery in the laboratory control sample (LCS) was below QC limits. Results for this analyte in associated samples may be biased low.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M6	Matrix spike and Matrix spike duplicate recovery not evaluated against control limits due to sample dilution.
N2	The lab does not hold NELAC/TNI accreditation for this parameter.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.
c2	Acid preservation may not be appropriate for the analysis of 2-Chloroethylvinyl ether.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 4610111

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
4610111001	Holding Tank 4	EPA 3545A	19824	EPA 8081B	19947
4610111001	Holding Tank 4	EPA 3510C	19712	EPA 8081B	19936
4610111001	Holding Tank 4	EPA 3545A	19573	EPA 8082A	19692
4610111001	Holding Tank 4	EPA 8151A	19826	EPA 8151A	19948
4610111001	Holding Tank 4	EPA 9056A	19902	EPA 9056A	20007
4610111001	Holding Tank 4	EPA 3050B	19198	EPA 6010C	19372
4610111001	Holding Tank 4	EPA 3010A	19511	EPA 6010C	19850
4610111001	Holding Tank 4	EPA 3050B	19197	EPA 6020A	19353
4610111001	Holding Tank 4	EPA 7470A	19863	EPA 7470A	19928
4610111001	Holding Tank 4	EPA 7471B	19958	EPA 7471B	20077
4610111001	Holding Tank 4	EPA 3550C	19230	EPA 8270C	19568
4610111001	Holding Tank 4	EPA 3510C	19713	EPA 8270C	19937
4610111001	Holding Tank 4	EPA 8260B	19532		
4610111002	Holding Tank 4	EPA 8260B	19961		
4610111001	Holding Tank 4	SM 2540 G-11/3550	19256		
4610111001	Holding Tank 4	SM 2540G	19254		
4610111001	Holding Tank 4	SM 2540G	19255		
4610111001	Holding Tank 4	EPA 9045D	19277		
4610111001	Holding Tank 4	EPA 351.2	19745	EPA 351.2	20083
4610111001	Holding Tank 4	SM 4500-NH3 B-11	19636	SM 4500-NH3 G-11	20224
4610111001	Holding Tank 4	SM 4500-P B-11	19313	SM 4500-P E-11	19498

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WO#: 4610111



4610111

AIN-OF-CUSTODY / Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed accurately.

P710 L12

Section A
Required Client Information:
Company: City of Cadillac
Address: 200 North Lake St.
Cadillac, MI 49601
Email: ctomaszewski@cadillac-mi.net
Phone: (231)775-2368
Requested Due Date:
Report To: Cindy Tomaszewski
Copy To:
Company Name:
Address:
Purchase Order #:
Project Name: Biosolids
Project #:
Section C
Invoice Information:
Attention:
Pace Project Manager: andrea.grube@pacelabs.com
Pace Profile #:
Regulatory Agency: MDEP
State / Location: MI

ITEM #	MATRIX	CODE	COLLECTED		DATE	TIME	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	SAMPLE CONDITIONS	Received on	Ice	Sealed	Custody	Cooler	Samples	Intact
			START	END															
1	HOLDING TANK 4				3/28/18	1340													
2																			
3																			
4																			
5																			
6																			
7																			
8																			
9																			
10																			
11																			
12																			

ADDITIONAL COMMENTS
(PP) = priority pollutants
Mtds. & RL same as for (2017)
KAR Project # 703913

RELINQUISHED BY / AFFILIATION: City of Cadillac
DATE: 3/28/18
TIME: 1420
ACCEPTED BY / AFFILIATION: [Signature]
DATE: 3/29/18
TIME: 1240

SAMPLE ID: One Character per box. (A-Z, 0-9 /, -)
Sample IDs must be unique

ANALYSES TEST Y/N
TCLP & RCRA Metals: X
TCLP & SVOC (PP): X
TCLP (PP): X
TCLP Volatiles: X
TCLP Pest/Herb: X
Residual Chlorine (Y/N): N

TEMP in C: 3.28
DATE Signed: 3.28.18

SAMPLER NAME AND SIGNATURE: [Signature]
PRINT Name of SAMPLER: DAVID S. H. F.
SIGNATURE of SAMPLER: [Signature]



SAMPLE RECEIVING / LOG-IN CHECKLIST

Client <i>City of Cadillac</i>	Work Order #: <i>4610111</i>
Receipt Record Page/Line # <i>19-20</i>	New / Add To Project Chemist Sample #s <i>001</i>

Recorded by (initials/date) <i>PS 3/29/18</i>	☒ Cooler ☐ Box ☐ Other	Qty Received <i>1</i>	☒ IR Gun (#202) ☐ Digital Thermometer (#54) ☐ Other (#)	☐ See Additional Cooler Information Form
--	--	--------------------------	--	---

Cooler #	Time	Cooler #	Time	Cooler #	Time	Cooler #	Time	
<i>Newy</i>	<i>1507</i>							
Custody Seals: ☒ None ☐ Present / Intact ☐ Present / Not Intact		Custody Seals: ☐ None ☐ Present / Intact ☐ Present / Not Intact		Custody Seals: ☐ None ☐ Present / Intact ☐ Present / Not Intact		Custody Seals: ☐ None ☐ Present / Intact ☐ Present / Not Intact		
Coolant Type: ☐ Loose Ice ☒ Bagged Ice ☐ Blue Ice ☐ None		Coolant Type: ☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		Coolant Type: ☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		Coolant Type: ☐ Loose Ice ☐ Bagged Ice ☐ Blue Ice ☐ None		
Coolant Location: Dispersed / Top / <u>Middle</u> / Bottom		Coolant Location: Dispersed / Top / Middle / Bottom		Coolant Location: Dispersed / Top / Middle / Bottom		Coolant Location: Dispersed / Top / Middle / Bottom		
Temp Blank Present: ☐ Yes ☒ No		Temp Blank Present: ☐ Yes ☐ No		Temp Blank Present: ☐ Yes ☐ No		Temp Blank Present: ☐ Yes ☐ No		
If Present, Temperature Blank Location is: ☐ Representative ☐ Not Representative		If Present, Temperature Blank Location is: ☐ Representative ☐ Not Representative		If Present, Temperature Blank Location is: ☐ Representative ☐ Not Representative		If Present, Temperature Blank Location is: ☐ Representative ☐ Not Representative		
Observed °C	Correction Factor °C	Actual °C	Observed °C	Correction Factor °C	Actual °C	Observed °C	Correction Factor °C	
Temp Blank:			Temp Blank:			Temp Blank:		
Sample 1:	<i>1.9</i>	<i>1.9</i>	Sample 1:			Sample 1:		
Sample 2:	<i>2.7</i>	<i>2.7</i>	Sample 2:			Sample 2:		
Sample 3:	<i>2.1</i>	<i>2.1</i>	Sample 3:			Sample 3:		
3 Sample Average °C:			3 Sample Average °C:			3 Sample Average °C:		
☐ Cooler ID on COC?			☐ Cooler ID on COC?			☐ Cooler ID on COC?		
☐ VOC Trip Blank received?			☐ VOC Trip Blank received?			☐ VOC Trip Blank received?		

If any shaded areas checked, complete Sample Receiving Non-Conformance and/or Inventory Form

Paperwork Received

Yes	No
☒	☐
☐ Chain of Custody record(s)? If No, Initiated By _____	
Received for Lab Signed/Date/Time?	
☐ Shipping document?	
☐ Other _____	

COC Information

☒ Pace COC	☐ Other _____
COC ID Numbers:	

Check COC for Accuracy

Yes	No
☒	☐
☐ Analysis Requested?	
☒	☐
☐ Sample ID matches COC?	
☒	☐
☐ Sample Date and Time matches COC?	
☒	☐
☐ Container type completed on COC?	
☒	☐
☐ All container types indicated are received?	

Sample Condition Summary

N/A	Yes	No
☒	☐	☒
☐ Broken containers/lids?		
☒	☐	☒
☐ Missing or incomplete labels?		
☒	☐	☒
☐ Illegible information on labels?		
☒	☐	☒
☐ Low volume received?		
☒	☐	☒
☐ Inappropriate or non-Pace containers received?		
☒	☐	☒
☐ VOC vials / TOX containers have headspace?		
☒	☐	☒
☐ Extra sample locations / containers not listed on COC?		

Check Sample Preservation

N/A	Yes	No
☒	☐	☒
☐ Temperature Blank OR average sample temperature, $\geq 6^{\circ}\text{C}$?		
☒	☐	☐
☐ If either is $\geq 6^{\circ}\text{C}$, was thermal preservation required?		
☒	☐	☐
If "Yes", Project Chemist Approval Initials: _____		
☒	☐	☐
If "Yes" Completed Non Con Cooler - Cont Inventory Form?		
☒	☐	☐
Completed Sample Preservation Verification Form?		
☒	☐	☐
☐ Samples chemically preserved correctly?		
☒	☐	☐
If "No", added orange tag?		
☒	☐	☐
☐ Received pre-preserved VOC soils?		
	☐ MeOH	☐ Na ₂ SO ₄

Check for Short Hold-Time Prep/Analyses

☐ Bacteriological	AFTER HOURS ONLY: COPIES OF COC TO LAB AREA(S) ☒ NONE RECEIVED ☐ RECEIVED, COCs TO LAB(S)
☐ Air Bags	
☐ EnCores / Methanol Pre-Preserved	
☐ Formaldehyde/Aldehyde	
☐ Green-tagged containers	
☐ Yellow/White-tagged 1 L ambers (SV Prep-Lab)	

Notes

<i>Biosolids more aqueous than solids</i>		
☐ Trip Blank received	☐ Trip Blank not listed on COC	
Cooler Received (Date/Time)	Paperwork Delivered (Date/Time)	≤ 1 Hour Goal Met?
<i>PS 3/29/18</i>	<i>PS 3/29/18</i>	Yes / No